

A large, circular image of Jupiter's south pole, showing a complex pattern of white and brown clouds and vortices. The image is centered on the slide, with text overlaid on it.

Lecture 11: Gas & Ice Giants

Jupiter, Saturn, Uranus, Neptune

Planetary Systems

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2026

Learning Objectives

By the end of this lecture, you will be able to:

- Compare the internal structures of the four giant planets.
- Describe atmospheric dynamics on Jupiter, Saturn, Uranus, and Neptune.
- Outline the diversity of giant-planet satellites and ring systems.
- Derive the Roche limit and apply it to Saturn's rings.
- Use the gas-giant / ice-giant dichotomy as a laboratory for exoplanet analogues.

Four giant planets, two physical classes, hundreds of moons, and only one mission ever to either ice giant.

Recap: Where We Are in the Course

- L9 (Earth & Venus): rocky-planet near-twins with divergent surface evolutions.
- L10 (Mercury & Mars): rocky-planet limiting cases (smallest/densest vs. low-mass volatile-rich).
- Today: the four giant planets, >99.5% of all planetary mass beyond the Sun.
- L12: small bodies; L13: exoplanets; L14: synthesis.



Two Classes, Four Planets

- **Gas giants:** Jupiter, Saturn
 - H/He envelopes dominate mass
 - 318 and 95 $M_{\oplus}$
 - Together: ~92% of all planetary mass
- **Ice giants:** Uranus, Neptune
 - H/He only ~10–20% of mass
 - Bulk = water, ammonia, methane fluids
 - 14.5 and 17.1 $M_{\oplus}$
- No solid surface; “1 bar level” is convention.
- Smooth gas → supercritical fluid (no liquid/gas distinction) → metallic plasma.
- Rapid rotation: ~ 10 h, oblate by 6–10%.

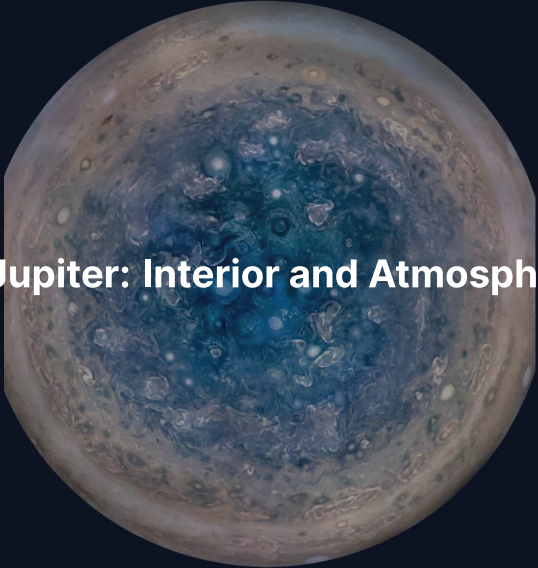
Energy Budgets: Internal Heat Excess

- Jupiter: emits $\sim 1.7\times$ absorbed solar flux $\rightarrow$ Kelvin-Helmholtz contraction.
- Saturn: $\sim 1.8\times \rightarrow$ helium-rain release of gravitational energy.
- Uranus: $\sim 1.06\times \rightarrow$ *anomalously low*; the major outlier.
- Neptune: $\sim 2.6\times \rightarrow$ source still uncertain.

Three of four giants are net energy exporters; Uranus is the puzzle that drives mission concepts.

Why Rapid Rotation Matters

- Jupiter: 9 h 56 min; Saturn: 10 h 33 min; rocky planets at 24 h or longer.
- Strong Coriolis → latitude-aligned bands of zonal wind.
- Oblateness encodes interior density via gravity harmonics $J_2, J_4, J_6, \dots$ (coefficients for departures from a sphere).
- Juno (Jupiter) and Cassini Grand Finale (Saturn) measured J_n to high order.
- Equatorial jet speeds counterintuitively *increase* with distance from Sun.

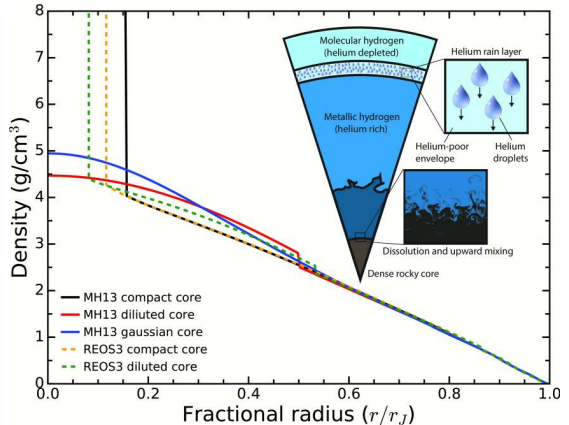
A circular image showing the south pole of Jupiter. The center is a deep blue, swirling vortex. Surrounding this are concentric rings of lighter blue and white, with some brownish spots. The outer edge of the image shows the brown and white bands of Jupiter's atmosphere.

2. Jupiter: Interior and Atmosphere

Jupiter's south pole, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS

Jupiter Interior: Layered Fluid

- Outer molecular H_2 + He envelope
- Transition to metallic hydrogen at ~ 100 GPa, ~ 5000 K
- $r/R_J \approx 0.85$ for the conducting transition
- Dynamo source (field-generating fluid motion): metallic hydrogen
- Centre: ~ 4000 GPa, $\sim 20\,000$ K



Wahl et al. (2017)

The Dilute Core: A Decade-Old Revolution

- Pre-Juno: assumed compact, well-defined $\sim 10 M_{\oplus}$ core.
- Juno gravity (J_4, J_6, J_8, J_{10}) cannot match compact-core models.
- Juno data fit naturally with heavy elements distributed across the inner 30–50% of the planet.
- "Dilute" / "fuzzy" core picture: Wahl+ 2017, Militzer+ 2022.
- Total heavy-element content: $\sim 20\text{--}40 M_{\oplus}$.

Dilute core implies efficient mixing or non-standard accretion; rewrites textbook formation picture.

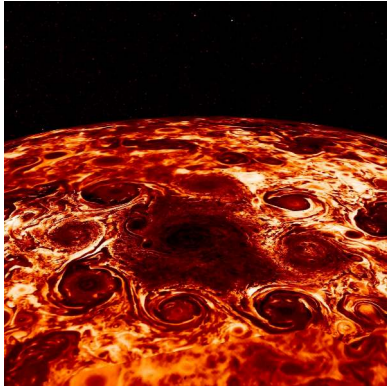


Crescent Jupiter and Great Red Spot, JunoCam (perijove 3, Dec 2016). Credit:
NASA/JPL-Caltech/SwRI/MSSS / R. Tkachenko

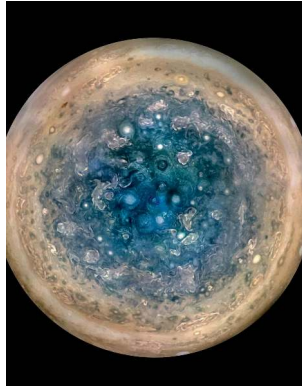
Reading the Jupiter Atmosphere Image

- Cloud decks at 0.1–10 bar: NH_3 (top), NH_4SH , H_2O (deepest).
- ~15 alternating zonal jets pole-to-pole; peak $\sim 180 \text{ m s}^{-1}$.
- Bright zones = rising air, ammonia clouds; dark belts = descending haze.
- Great Red Spot: anticyclonic (counter-rotating) vortex observed since at least 1830.
- GRS shrinking: $\sim 40\,000 \text{ km}$ in 1900 $\rightarrow \sim 14\,000 \text{ km}$ today.
- Galileo probe (1995): confirmed layering but found a dry entry site.

Polar Cyclones



North pole: 1 + 8 cyclones, Adriani+ 2018



South pole: 1 + 5 cyclones

Stable polygonal arrangements persist across many Juno orbits. Polar weather not seen at high resolution before 2016.

Jets, Aurorae, and the Great Blue Spot

- Kaspi+ 2018: equatorial jets penetrate to ~3000 km depth (gravity asymmetries).
- Below: magnetic stress damps differential rotation; deep interior → solid-body.
- Aurorae are the strongest in the solar system.
- Driven by the magnetosphere (planet's magnetic-field bubble) + Io mass loading ($\sim 1 \text{ t s}^{-1}$ of S, O).
- Auroral footprints of Io, Europa, Ganymede directly imaged in UV.
- "Great Blue Spot": isolated equatorial magnetic anomaly mapped by Juno (Connerney+ 2022).

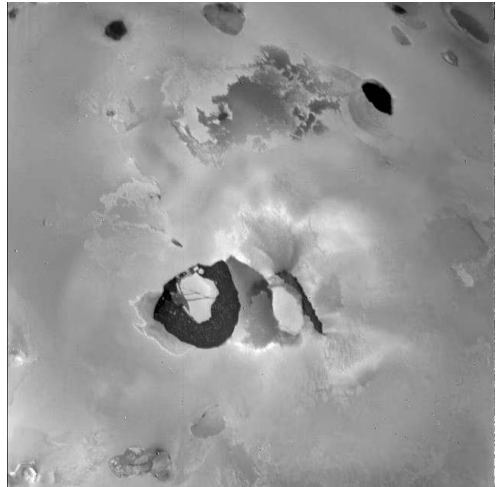
3. The Galilean Moons



The Galilean moons at correct relative size: Io, Europa, Ganymede, Callisto. Credit: NASA/JPL/DLR

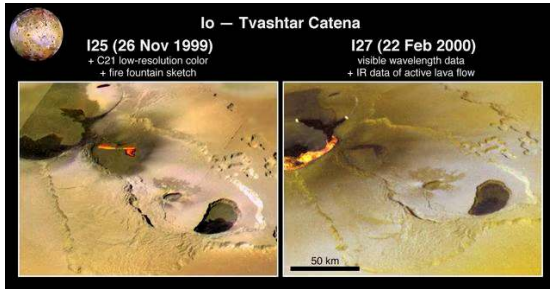
Io: Most Volcanic Body in the Solar System

- ~400 active volcanic centres
- Heat output $\sim 10^{14}$ W (tidal, not radiogenic)
- Driven by 1:2:4 Laplace resonance (orbital periods locked in integer ratios) with Europa, Ganymede
- Predicted by Peale+ 1979, days before Voyager 1
- Surface resurfaced on ~Myr timescales



Loki Patera lava lake, Voyager 1 (1979).
NASA/JPL

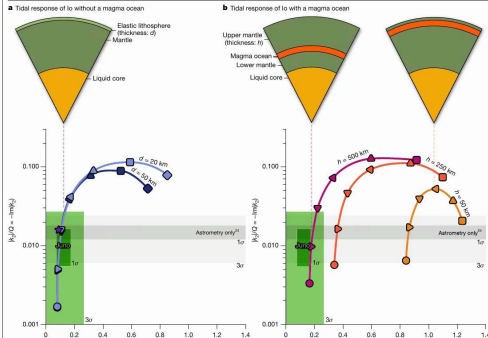
Io: Tvashtar Eruption



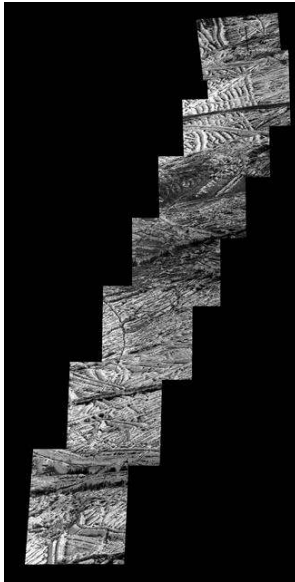
- Tvashtar Catena fire fountain imaged by Galileo (orbits I25, Nov 1999 and I27, Feb 2000).
- Composite of low-resolution colour, visible imaging, and IR detection of active lava flow.
- Tenuous SO_2 atmosphere: sublimates by day, freezes at night.
- Strong day-night asymmetries.

Io Interior: Juno Says No Magma Ocean

Article



- Juno close flybys 2023–2024.
- Two-way Doppler → tidal Love number k_2 (deformability under tides) $\approx 0.125 \pm 0.047$.
- **Too small** for global shallow magma ocean.
- Mostly solid silicate mantle with localised partial melts.
- Park et al. (2024).

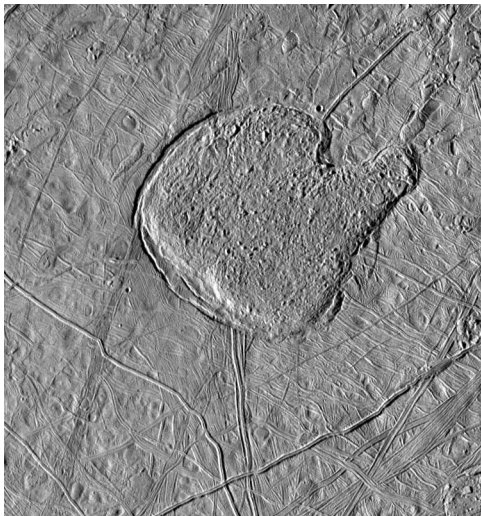


Europa surface, Galileo orbiter. Credit: NASA/JPL-Caltech/SETI Institute

Reading Europa: Lineae and Chaos

- Surface dominated by water ice; criss-crossed by long fractures (lineae).
- Patches of "chaos terrain": ice has foundered, broken up, refrozen.
- Crater counts: surface age only 40–90 Myr (Pappalardo+ 1999).
- Diameter 3122 km (smaller than Moon).
- Strong tidal heating sustained by Laplace resonance with Io and Ganymede.

The Case for Europa's Ocean



Conamara Chaos, Galileo PIA01640

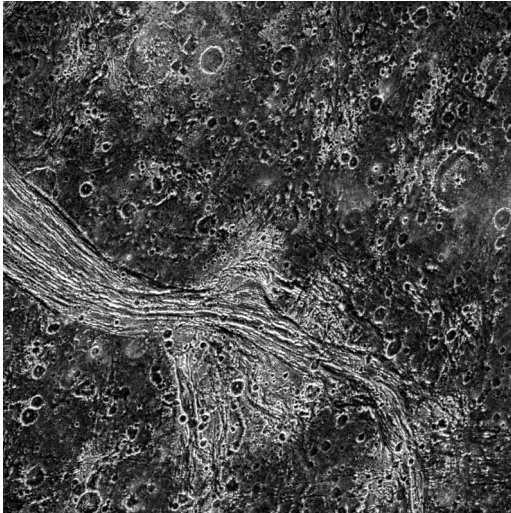
- Chaos morphology → episodic basal melting.
- Galileo magnetometer: induced field requires global conductor (Khurana+ 1998).
- Hubble UV: candidate plumes near south pole (Roth+ 2014, Sparks+ 2017).
- Modern picture: 15–25 km ice over ~100 km ocean, ~Earth-ocean volume.

Ganymede: Largest Moon, Only Moon-Dynamo



- Radius 2634 km: bigger than Mercury.
- Only moon known to host an intrinsic dynamo (Galileo, 1996).
- Fully differentiated: Fe core → silicate mantle → icy shell.
- Ocean: ~100 km deep below ~150 km ice (Saur+ 2015 from HST aurorae).

Ganymede: Grooved Terrain



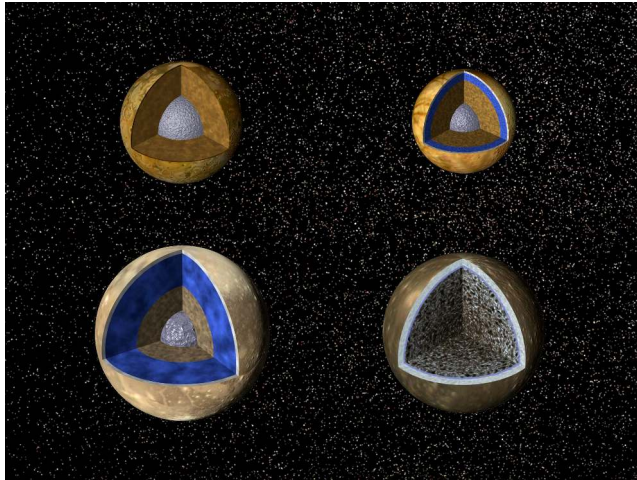
- Lagash Sulcus: bright grooved terrain cuts dark cratered terrain.
- Records ancient tectonic extension.
- Distinct from Callisto's intact, undisturbed surface.
- ESA's JUICE will orbit Ganymede in 2034.

Callisto: The Quiet Sibling



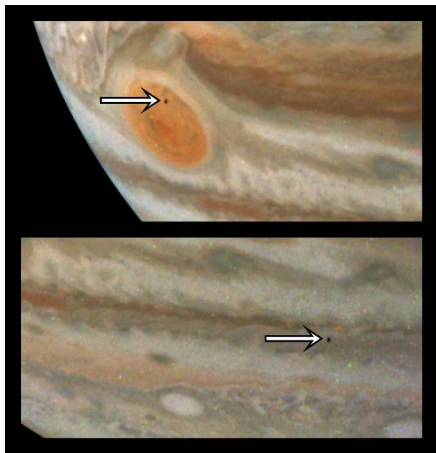
- $R = 2410 \text{ km}$, density 1834 kg m^{-3} .
- $C/MR^2 \approx 0.355$ (moment of inertia; lower = more centrally concentrated): only *partially* differentiated (Anderson+ 2001).
- No tectonic resurfacing; outside the 1:2:4 Laplace lock.
- Yet: induced field signal $\rightarrow$ probable subsurface ocean.
- Lower radiation than inner Galileans ("boring but habitable").

Galilean Comparative Anatomy



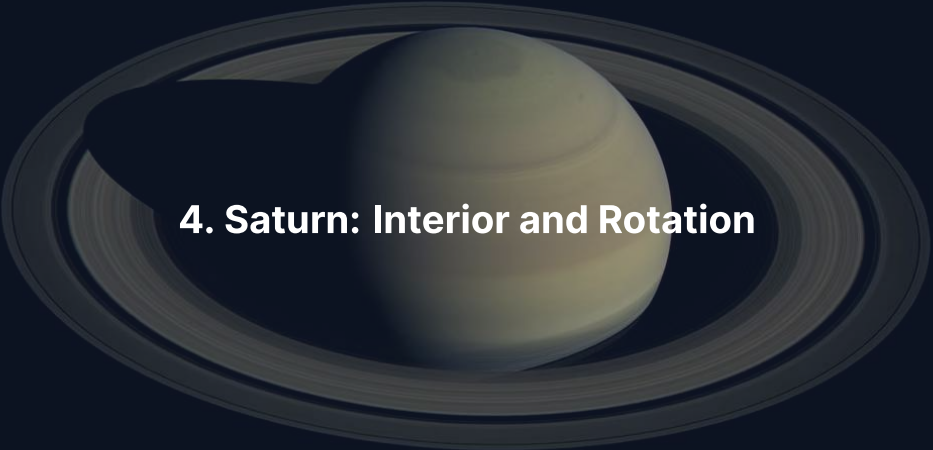
Interior structures (l-r: Io, Europa, Ganymede, Callisto). NASA/JPL artist concept (PIA01082).

Jupiter's Rings and Small Moons



Amalthea silhouette against Jupiter, Juno PJ59 (2024).

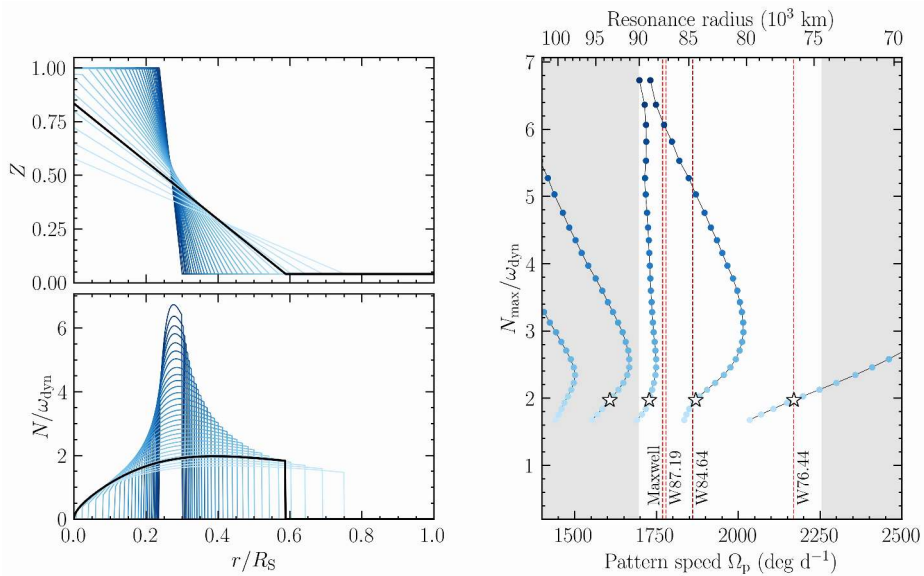
- Faint dusty rings, discovered by Voyager 1 (1979).
- Sourced by impact gardening of Amalthea, Adrastea, Metis, Thebe.
- Continuously replenished, transient → *not* a long-lived disk.

A photograph of Saturn and its rings, taken by the Cassini spacecraft. The planet is a pale yellowish-white sphere with a dark shadow cast to its left. The rings are a complex system of many thin, overlapping rings, appearing as a series of concentric ellipses. The background is a dark, deep blue.

4. Saturn: Interior and Rotation

Saturn Interior: Lower Pressures, Helium Rain

- Mass $95 M_{\oplus}$; central pressure $\sim 1/3$ Jupiter's.
- Molecular-metallic hydrogen transition deeper, milder conditions.
- Cassini Grand Finale (2017): gravity field rivalling Juno's at Jupiter.
- Helium becomes immiscible in metallic H at 1–3 Mbar, 5–10 kK.
- Droplets fall, release gravitational PE $\rightarrow$ closes Saturn's energy budget.
- Predicts He depletion in upper envelope; consistent with Voyager.



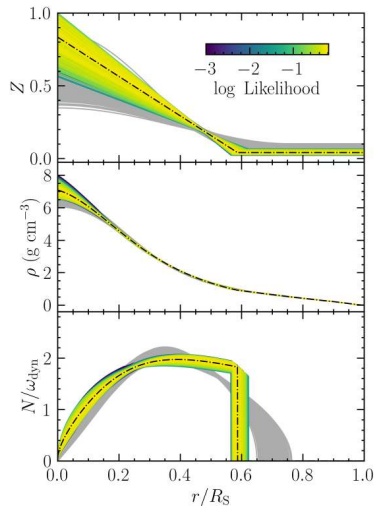
Kronoseismology constraints on Saturn's interior, Mankovich & Fuller (2021).

Reading the Kronoseismology Result

- Saturn's f-mode oscillations excite density waves in the C ring at resonance radii.
- Pattern speeds matched to wave positions in 75 000–95 000 km range.
- Match requires *stably stratified, dilute* heavy elements out to $\sim 60\%$ of R_S .
- Total heavy-element mass $\sim 17 M_\oplus$.
- First seismic constraint on *any* giant planet interior.

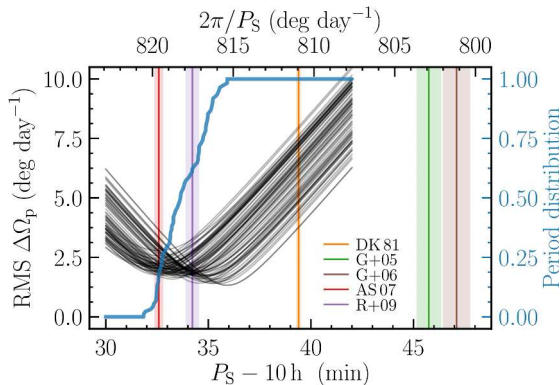
Like Jupiter, Saturn has a fuzzy core, not a compact one. Both giants share the same surprise.

Heavy-Element Distribution Profile



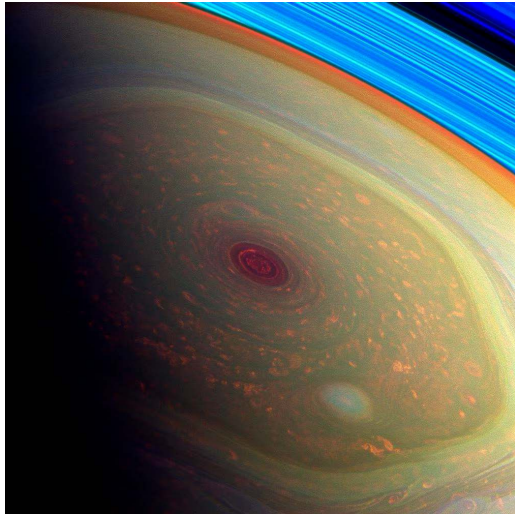
- Top: $Z(r)$ (heavy-element mass fraction).
- Middle: density $\rho(r)$.
- Bottom: Brunt-Väisälä frequency N .
- Yellow: maximum-likelihood profile.
- Mankovich & Fuller (2021).

Saturn's Rotation Period from Ring Seismology



- Saturn's magnetic dipole nearly axisymmetric $\rightarrow$ no clean radio period.
- C-ring f-mode wave residuals as a function of assumed P .
- Best estimate: $P_s = 10 \text{ h } 33 \text{ min } 38 \text{ s}$.
- Mankovich+ 2019.

Saturn's Atmosphere and the Hexagon



- Same $\text{NH}_3/\text{NH}_4\text{SH}/\text{H}_2\text{O}$ cloud decks as Jupiter, deeper and stretched.
- Banded contrast muted in visible.
- Hexagonal jet at $\sim 78^\circ \text{N}$: standing Rossby wave (large-scale atmospheric wave), six sides.
- Continuous since Voyager (1981).
- Equatorial jet $\sim 400 \text{ m s}^{-1}$ (Cassini).

A photograph of Saturn and its rings, taken from the Cassini spacecraft. The planet is a pale yellowish-white sphere with a dark shadow cast to its left. The rings are a complex system of many thin, overlapping rings, appearing as a series of concentric ellipses. The background is a dark, deep blue.

5. Saturn's Rings

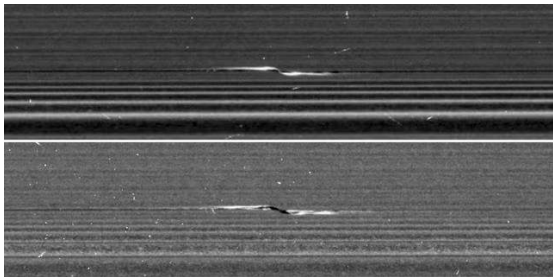


Natural-colour radial scan: C, B ring, Cassini Division, A ring. Cassini (PIA08389).
NASA/JPL-Caltech/SSI

Reading the Ring Scan

- Main ring system spans 67 000–140 000 km from Saturn's centre.
- D, C, B rings inside the fluid Roche limit (computed below: ~126 000 km).
- Cassini Division at ~118 000 km: cleared by 2:1 resonance with Mimas.
- Outer A ring + F ring: just outside Roche; A-ring edge sharpened by 7:6 with Janus.
- Composition: >95% water ice, particle sizes 1 cm to 10 m.
- Vertical extent only ~10 m: thinnest known objects in the universe (relative to extent).

Embedded Moons and Propellers



- F ring shepherded (edges confined gravitationally) by Prometheus + Pandora.
- Encke Gap held open by embedded moon Pan.
- "Propeller" features: ~1 km moonlets too small to clear gaps.
- Direct measurement of orbital migration in a disk (Tiscareno+ 2013).

The Rings Are Young

- Total ring mass $\sim 1.5 \times 10^{19}$ kg = 40% of Mimas (less+ 2019, Cassini gravity).
- Far too small for primordial origin: would have darkened by infall over 4.5 Gyr.
- Brightness + mass + composition $\rightarrow$ age ~ 100 Myr (Crida+ 2019).
- "Ring rain": water + organics infall onto Saturn's atmosphere (Waite+ 2018).
- Ground-based H_3^+ ionosphere modulation confirms (O'Donoghue+ 2019).

Remaining lifetime $\sim 10^8$ yr; Saturn's rings are a transient phase, not a permanent feature.

What Made the Rings?

- Wisdom+ (2022): tidal disruption of an icy Mimas-class moon scattered inside the Roche limit.
- Could simultaneously explain ring origin and the unusual mid-sized moon orbits.
- Alternative: late capture of a Kuiper Belt object.
- Charnoz+ (2009): possibly older but with continuous mass shedding.
- Pedagogical message: solar-system bodies do not all date from 4.5 Gyr ago.

6. Blackboard Derivation: The Roche Limit



Setup: Why Rings Exist

- Roche limit: distance inside which a fluid satellite's self-gravity loses to the planet's tidal stretch.
- Concept introduced by Roche (1849).
- Sets where rings can exist and where moons cannot accrete.

Configuration:

- Planet: mass M_p , radius R_p , density ρ_p .
- Satellite: mass M_s , radius R_s , density ρ_s , at distance d from planet centre.
- Assume $d \gg R_s$ so a Taylor expansion across the satellite is valid.
- Work in the satellite's rest frame.

Step 1: Tidal Acceleration Across the Satellite

Acceleration at the satellite's centre:

$$g_{\text{centre}} = \frac{GM_p}{d^2}$$

At the near point (closest to planet, distance $d - R_s$):

$$g_{\text{near}} = \frac{GM_p}{(d - R_s)^2}$$

Differential (tidal) acceleration that stretches the satellite:

$$\Delta a_{\text{tidal}} = g_{\text{near}} - g_{\text{centre}} = GM_p \left[\frac{1}{(d - R_s)^2} - \frac{1}{d^2} \right]$$

Step 2: Taylor Expand in R_s/d

For $R_s \ll d$:

$$\frac{1}{(d - R_s)^2} = \frac{1}{d^2} \left(1 - \frac{R_s}{d}\right)^{-2} \approx \frac{1}{d^2} \left(1 + \frac{2R_s}{d}\right)$$

So:

$$\Delta a_{\text{tidal}} \approx \frac{2GM_p R_s}{d^3}$$

Tidal forces fall off as d^{-3} , not d^{-2} . This is the universal scaling.

Step 3: Self-Gravity at the Satellite's Surface

Uniform density ρ_s :

$$a_{\text{self}} = \frac{GM_s}{R_s^2} = \frac{G \cdot \frac{4}{3}\pi R_s^3 \rho_s}{R_s^2} = \frac{4}{3}\pi G \rho_s R_s$$

Setting tidal = self-gravity at the disruption threshold:

$$\frac{2GM_p R_s}{d^3} = \frac{4}{3}\pi G \rho_s R_s$$

Step 4: Solve for d_R

Cancel G and R_s , substitute $M_p = \frac{4}{3}\pi R_p^3 \rho_p$:

$$\frac{2 \cdot \frac{4}{3}\pi R_p^3 \rho_p}{d^3} = \frac{4}{3}\pi \rho_s$$

Rearrange:

$$d^3 = 2R_p^3 \frac{\rho_p}{\rho_s}$$

$$d_R^{\text{rigid}} = 2^{1/3} R_p \left(\frac{\rho_p}{\rho_s} \right)^{1/3} \approx 1.26 R_p \left(\frac{\rho_p}{\rho_s} \right)^{1/3}$$

Fluid vs. Rigid Roche Limit

- Rigid case: spherical, holds shape until disruption.
- Fluid case: deforms into ellipsoid first, disrupts further out.
- Roche's full fluid calculation gives prefactor ≈ 2.46 :

$$d_R \approx 2.46 R_p \left(\frac{\rho_p}{\rho_s} \right)^{1/3}$$

Real bodies with finite material strength sit between the two limits.

Step 5: Apply to Saturn's Rings

$$R_p = 58,232 \text{ km}, \quad \rho_p = 687 \text{ kg m}^{-3}, \quad \rho_s \approx 1000 \text{ kg m}^{-3}$$

$$d_R \approx 2.46 \times 58,232 \times (0.687)^{1/3} \text{ km} \approx 2.46 \times 58,232 \times 0.883$$

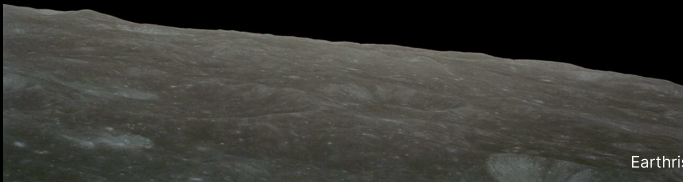
$$d_R \approx 126,000 \text{ km}$$

Within ~10% of A-ring outer edge (~137 000 km). Discrepancy: ring-particle porosity, ice rigidity, density-wave structure.

Why Roche Matters

- Inside d_R : differential pull > self-gravity → no agglomerate > 10 m holds together.
- Particles orbit as a collisional disk; cannot accrete into a moon.
- Outside d_R : same population can re-accrete (e.g., small inner moons of Saturn).
- Used: Saturn ring origin, Triton's eventual fate, comet Shoemaker-Levy 9 disruption (Jupiter, 1992).

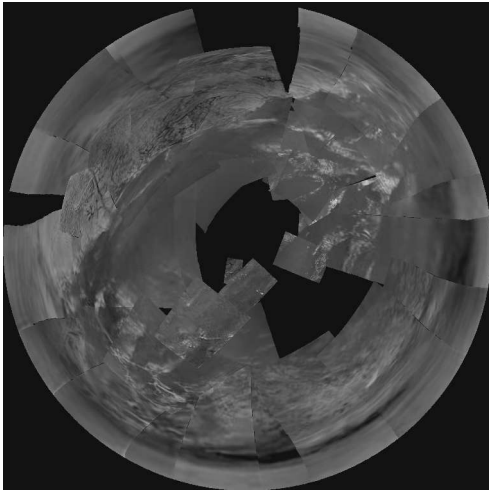
Break





7. Saturn's Moons: Titan, Enceladus, and More

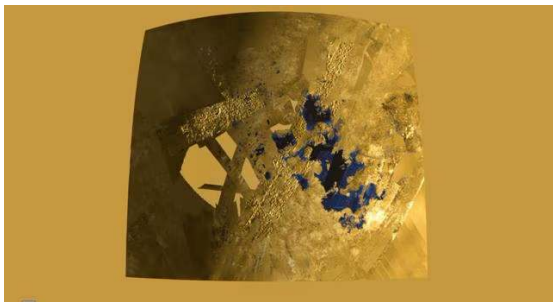
Titan: Atmosphere and Methane Cycle



Huygens descent imagery, 14 Jan 2005
(PIA07870)

- $R = 2575$ km; second largest moon.
- N_2 atmosphere: 1.5 bar surface, $4\times$ Earth column density.
- Surface 94 K $\rightarrow$ methane near triple point.
- Methane: rain, rivers, lakes, evaporation cycle.

Titan: Polar Lakes

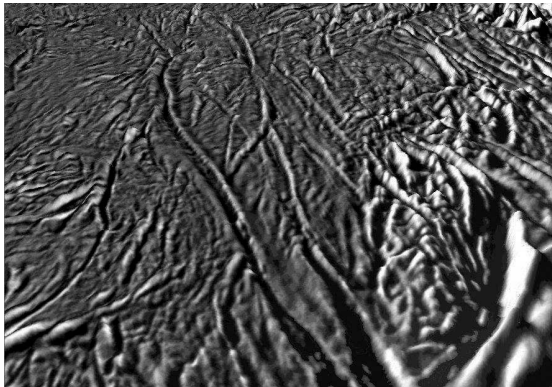


- Hundreds of lakes near the north pole (Cassini RADAR).
- Kraken Mare, Ligeia Mare ~Caspian-sized.
- Photochemistry produces "tholins": HCN , C_2H_2 , ..., organic haze.
- Tholins settle → form equatorial dunes.

Titan: Subsurface Ocean

- Cassini gravity + tidal Love number → global subsurface water-ammonia ocean (less+ 2012).
- Ice shell ~100 km thick.
- Joins Europa, Ganymede, Callisto, Enceladus on the ocean-worlds list.
- NASA's **Dragonfly** (launch 2028, arrival 2034): first rotorcraft on another world.
- Targets Selk impact crater for organics + transient melt-water chemistry.

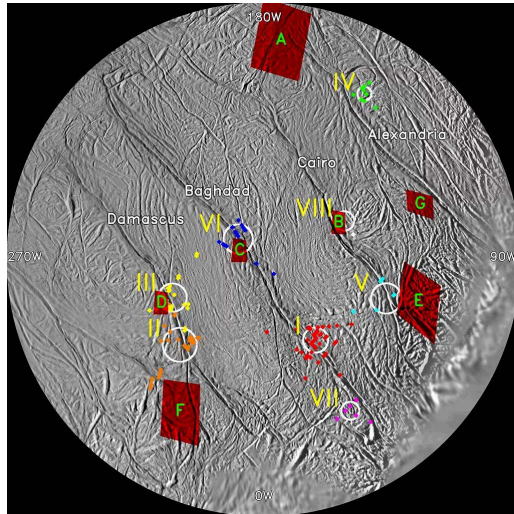
Enceladus: A Moon That Erupts



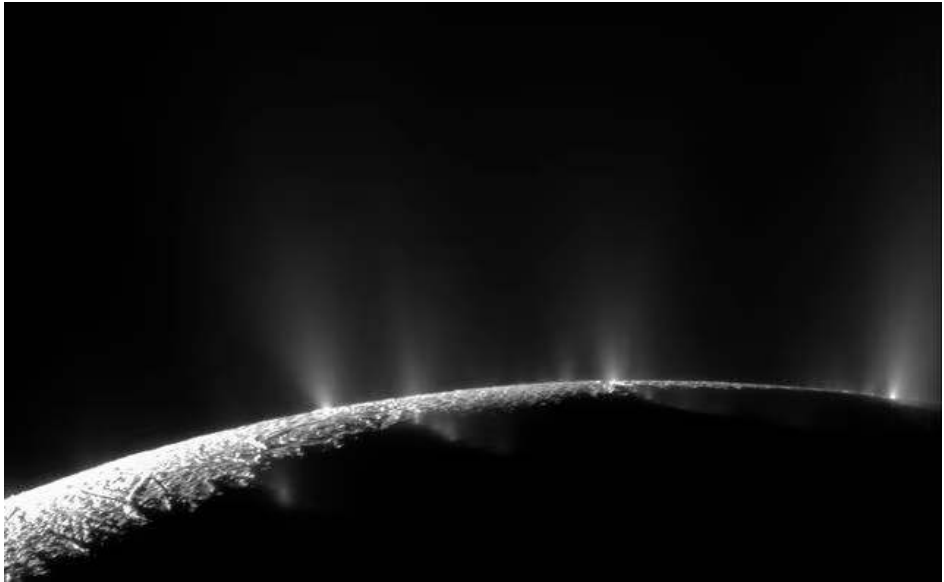
Tiger stripes (south polar fractures)

- $R = 252$ km, but oversized scientific impact.
- 2005 Cassini discovery: active geyser plumes from south-pole "tiger stripes".
- Porco+ 2006.
- Plume composition: 90% H_2O , also CO_2 , NH_3 , salts, organics.

Enceladus: Heat at the Stripes



- Cassini CIRS: stripes 10s of K warmer than surrounding terrain.
- Jet sources align with hot spots.
- Total south-polar heat > 10 GW, far above radiogenic budget.
- Sustained by 2:1 resonance with Dione.



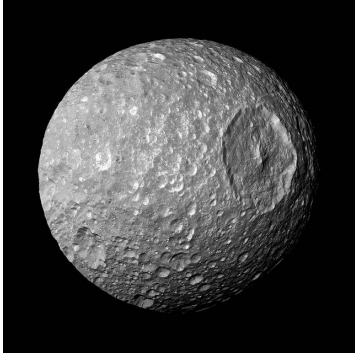
Dozens of geyser jets erupting from Enceladus's tiger stripes. Cassini, NASA/JPL-Caltech/SSI.

Enceladus: Habitability Ingredients

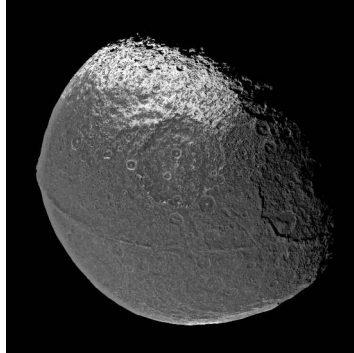
- Global ocean below ~20–30 km ice shell.
- Plume H_2 (Waite+ 2017) → thermodynamic disequilibrium, hydrothermal-vent analogue.
- Silica nanoparticles (Hsu+ 2015) → rock-water reactions today.
- Phosphates (Postberg+ 2023) → key biochemistry building block in seawater.
- Cassini's mass spectrometers were not designed to detect biosignatures.
- 2022 Decadal Survey priority: future Enceladus orbilander or sample return.

Most accessible ocean-world habitability target; plumes deliver the ocean to your spacecraft.

Mid-Sized Moons: Mimas and Iapetus

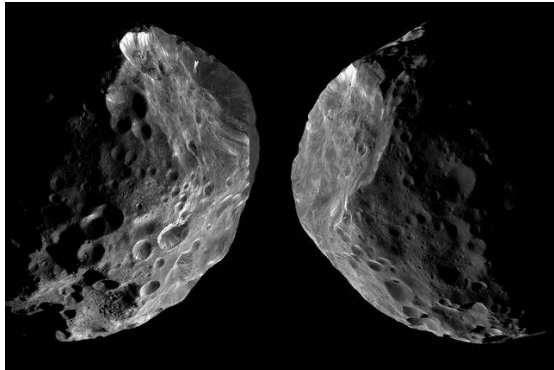


Mimas, Herschel crater (close to disruption threshold)



Iapetus, two-toned hemispheres

Phoebe: A Captured Kuiper Belt Object

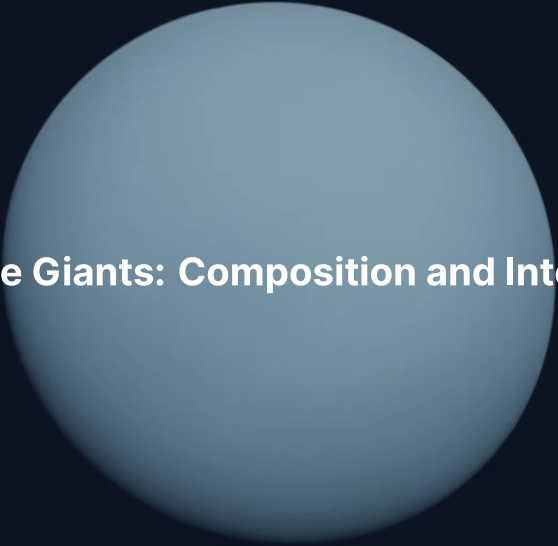


- Outer irregular moon (likely captured, not formed in place), retrograde, highly inclined.
- Colour, density, surface chemistry → captured KBO (Agnor & Hamilton 2006).
- Source of dark dust on Iapetus's leading hemisphere.
- A free trans-Neptunian sample, accessible without leaving Saturn.

Break

End of Part 1 (Gas Giants)

Part 2: Ice Giants Uranus and Neptune



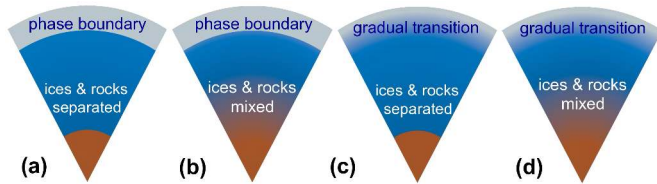
8. Ice Giants: Composition and Interior

Ice Giants: The Exotic Twins

- Uranus: $14.5 M_{\oplus}$, $4.0 R_{\oplus}$.
- Neptune: $17.1 M_{\oplus}$, $3.9 R_{\oplus}$.
- H/He envelope only 10–20% of mass.
- Bulk: water, ammonia, methane in compressed-fluid form.
- "Ice" reflects formation as solid grains; current state is fluid.
- One spacecraft each: Voyager 2 in Jan 1986 / Aug 1989. No return mission since.

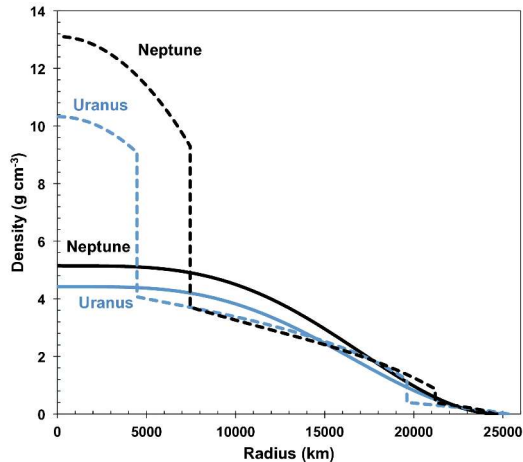
Most under-explored major planets. Knowledge ~35–40 yr out of date.

Possible Interior Structures



- (a) Sharp boundaries.
- (b) Sharp envelope, gradual ice/rock.
- (c) Gradual envelope, sharp ice/rock.
- (d) Fully gradual gradient.
- Voyager J_2, J_4 alone cannot distinguish.
- Helled+ (2020).

Density Profile Degeneracy



- Uranus (blue) and Neptune (black).
- Solid: empirical; dashed: 3-layer.
- Both fit Voyager gravity equally well.
- Cannot uniquely solve interior with current data.
- Motivates a dedicated orbiter.

Superionic Ice

- Millot+ (2019): laser shock to 100–400 GPa, several thousand K.
- In-situ X-ray diffraction → rigid bcc (body-centred cubic) oxygen lattice with mobile protons.
- Material is part solid, part liquid; *ionically* conducting.
- Can sustain a planetary dynamo without metallic hydrogen.
- Soderlund+ (2020): thin-shell dynamo → multipolar, off-axis fields.

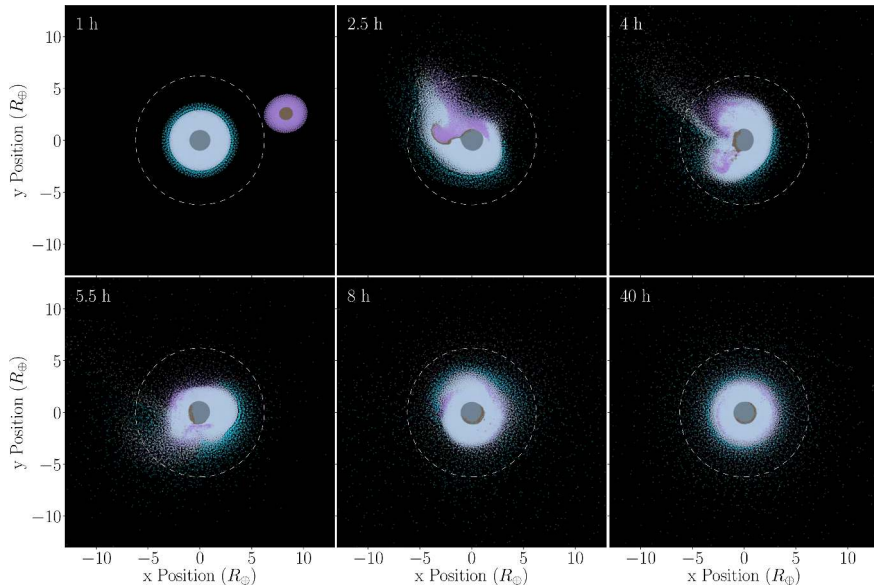
Superionic ice is the leading dynamo source for Uranus and Neptune.

Ice Giant Magnetic Fields

- Uranus: dipole tilt $\sim 59^\circ$ from rotation axis, offset $\sim 1/3R_p$ from centre.
- Neptune: dipole tilt $\sim 47^\circ$, similarly offset.
- Strong quadrupole + higher-order components dominate.
- Very different from dipole-dominated Jupiter and Saturn.
- Connerney+ 1991.
- Best explanation: thin convecting shell of ionic / superionic fluid.



9. Uranus: The Tilted Planet

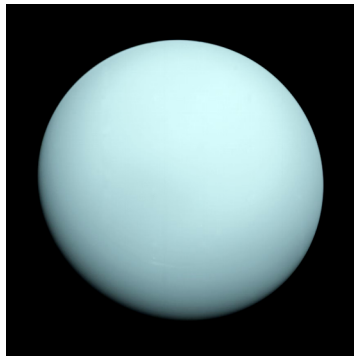


SPH simulation of giant impact on proto-Uranus. Kegerreis et al. (2018).

Reading the Uranus-Tilt Simulation

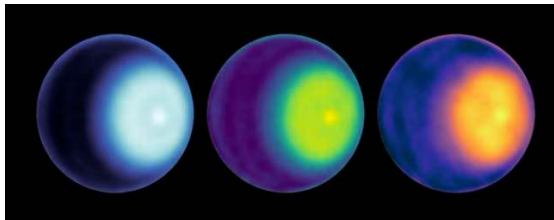
- Axial tilt 97.8° : rotation axis nearly in orbit plane.
- Each pole has 42 yr of daylight then 42 yr of night.
- Most plausible cause: oblique giant impact in late accretion.
- Snapshots ($t = 1$ to 40 h): light/dark grey = target ice/rock; blue = target H/He.
- Same impact must have produced equatorial debris disk that later re-accreted to satellites (Morbidelli+ 2012).

Uranus: Featureless at Voyager, Active Now



Voyager (1986) saw Uranus near solstice → pole-on, muted cloud bands.

Modern Uranus: Storms and Cyclones



- Approaching equinox: more cloud activity each year.
- 2014 storm system: amateur-detectable.
- JWST 2023 (De Pater+ 2022): vivid rings, polar cap structure.
- North-polar cyclone confirmed by VLA + JWST.

Uranus: Why So Cold Inside?

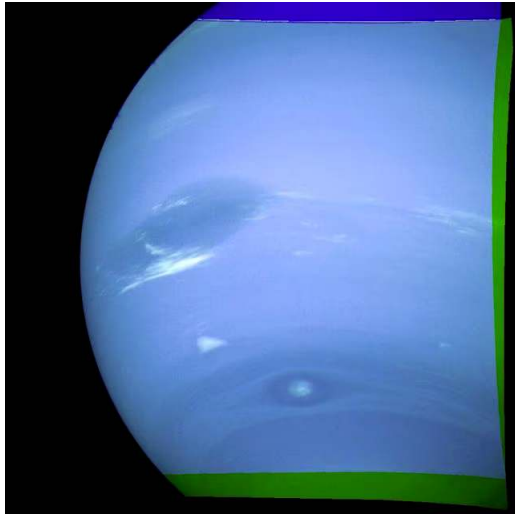
- Voyager IRIS: emits only $\sim 1.06\times$ absorbed solar flux (Pearl+ 1990).
- Neptune at similar distance: $\sim 2.6\times$. Why is Uranus so quiet?
- Hypothesis 1: giant impact left stably stratified, inhomogeneous interior; convection inhibited, primordial heat trapped.
- Hypothesis 2: latent heat from condensation of CH_4 or H_2O at depth masks deep flux.
- Hypothesis 3: external perturbation reset thermal history.
- Open question; demands dedicated orbiter.



10. Neptune and Triton

Neptune Great Dark Spot, Voyager 2 (1989). NASA/JPL-Caltech

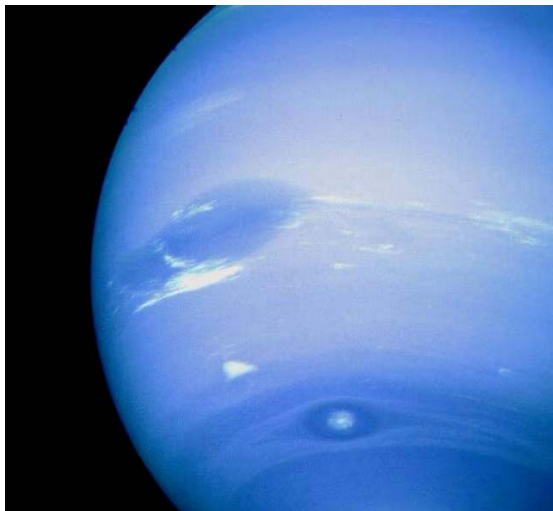
Neptune: The Active Ice Giant



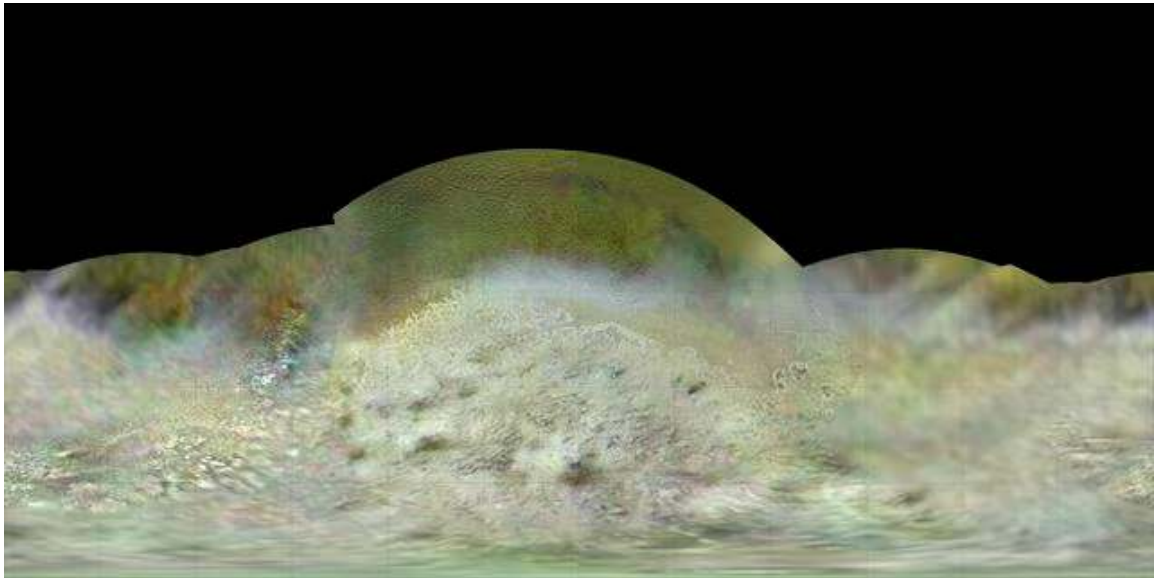
Great Dark Spot, Voyager 2 (1989)

- Active anticyclonic storm at flyby.
- GDS comparable in size to Earth.
- Bordered by bright methane cirrus.
- By 1994 (HST): GDS had vanished.
- Subsequent dark spots come and go on decade timescales.

Neptune: Fastest Winds in the Solar System



- "Scooter": small bright cloud at $\sim 380 \text{ m s}^{-1}$.
- Equatorial easterly jet peaks $\sim 580 \text{ m s}^{-1}$.
- Yet receives only 1/900 of Earth's insolation.
- Energy must come from interior.
- Internal flux $\sim 2.6\times$ absorbed (Pearl & Conrath 1991).
- Source mechanism: still unresolved.



Triton's southern hemisphere, Voyager 2 mosaic. NASA/JPL-Caltech.

Reading Triton

- Largest moon of Neptune; *retrograde* orbit ($i \approx 157^\circ$ relative to equator).
- Captured Kuiper Belt object: orbit cannot be primordial.
- Capture probably via tidal disruption of a binary near Neptune (Agnor & Hamilton 2006).
- Surface age $\lesssim 100$ Myr from crater counts $\rightarrow$ recent resurfacing.
- "Cantaloupe terrain" is unique in the solar system.
- Voyager observed 8 km tall N_2 geysers at the south pole.
- Will spiral into Roche limit on ~ 3.6 Gyr timescale.

A black and white photograph of the Neptune ring system, showing two distinct, slightly curved ring arcs against a dark, star-speckled background. The rings are composed of small particles, creating a grainy texture. The text "11. Ice Giant Rings" is overlaid in white, bold font in the center of the image.

11. Ice Giant Rings

Neptune ring system, Voyager 2 (1989). NASA/JPL-Caltech

Uranus Rings: Narrow, Dark, Discovered by Occultation

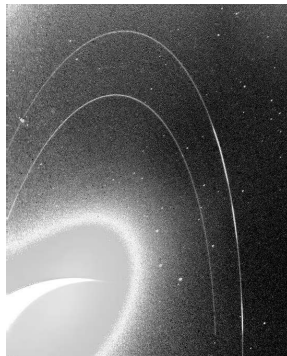
- Discovered 1977 from a stellar occultation: 5+5 dimming events.
- Voyager 2 (1986): 11 rings; HST (2003–2005): 13 total.
- Most prominent: ϵ ring at ~51 000 km.
- All narrow (≤ 10 km), dark, carbon-rich.
- Some confined by small "shepherd" moonlets.

Neptune Rings: Arcs Held by Resonance



Adams ring + Le Verrier ring

Adams ring contains 5 distinct *arcs*, gravitationally trapped by mean-motion resonance with Galatea.



Long-exposure forward-scattered view

Why the Compositional Difference?

- Saturn's rings: >95% bright water ice.
- Uranus + Neptune rings: dark, carbon-rich.
- Different formation pathway likely: disruption of small inner moons, not large external sources.
- JWST 2023: revealed previously undetected fine structure at Uranus.
- Brighter Adams ring arcs have visibly faded since Voyager.

A conceptual illustration of the Europa Clipper spacecraft in orbit around Jupiter. The spacecraft, featuring a central body and two large solar panel arrays, is positioned in the center-left of the frame. The curved, orange-brown horizon of Jupiter dominates the left side of the image, while the deep blue-black space of the background fills the rest. The title text is overlaid on the right side of the image.

12. Comparative Payoff and Exploration Frontier

Why Gas Giants and Ice Giants Diverged

- All four began as solid cores accreting disk gas (L2).
- For runaway gas accretion (envelope collapse once critical mass is reached): core $\gtrsim 10 M_{\oplus}$ *while gas is still present*.
- Disk lifetime 3–5 Myr is the deadline.
- Jupiter, Saturn: cores reached the threshold early enough.
- Uranus, Neptune: too late, or low-density region $\rightarrow$ stalled at 10–20 $M_{\oplus}$.
- Nice model (giant-planet migration scenario, Tsiganis+ 2005): ice giants migrated and scattered planetesimals (leftover km-scale bodies) after dispersal.

Ice-giant masses are a natural intermediate outcome of core accretion in the outer disk.

Common Themes Across All Four Giants

1. Internal heat excess (Uranus is the outlier).
2. Strong zonal jets; speed *increases* with distance from Sun.
3. Global magnetic fields, but very different morphologies (dipole vs. multipole).
4. Moon and ring systems, but different inventories: bright icy moons (Saturn) vs. captured KBOs (Triton).
5. All four are laboratories for sub-Neptune and Neptune-mass exoplanets (L13).

What We Still Don't Know

- Heavy-element fraction of Uranus and Neptune?
- What sustains Neptune's heat flow, and why is Uranus so quiet?
- How and when did Saturn's rings form? How long will they last?
- Origin of Jupiter's dilute core?
- Does Callisto have a true ocean today?
- Deep structure of Saturn's hexagonal jet?
- Habitability of Enceladus, Europa, and Titan?

Two Ocean-World Missions to Jupiter



Europa Clipper (NASA, launched Oct 2024)

- **Europa Clipper:** ~50 close flybys of Europa from Jovian orbit.
- REASON radar, magnetometer, mass spec, thermal imager.
- **JUICE** (ESA, launched Apr 2023): orbits Ganymede in 2034.
- First-ever orbiter of any moon other than Earth's.
- Complementary, not redundant.

Future: Dragonfly and a Uranus Orbiter

- Dragonfly (NASA, launch 2028, arrival 2034): first rotorcraft on another world.
- Selk crater on Titan: organics + transient melt-water from impact heat.
- Mass spec orders of magnitude better than Cassini's.
- 2022 Decadal Survey: **Uranus orbiter + probe** = top large flagship for the 2030s.
- Jupiter gravity assist → orbital insertion + atmospheric probe.
- Launch in early 2030s → arrival late 2030s / early 2040s.

The Voyager Legacy

- Launched 1977; only spacecraft ever to visit Uranus (1986) and Neptune (1989).
- Voyager 1: directed out of ecliptic after Saturn.
- Both still operational in interstellar space; expected loss of contact ~2030.
- Uranus and Neptune flyby data still yielding new science.
- Lesson 1: a single short flyby cannot substitute for sustained investigation.
- Lesson 2: well-designed missions return science long after the spacecraft is gone.

Summary: Today's Course-Level Takeaways

1. Gas giants and ice giants form a compositional continuum, set by formation timing.
2. Both gas giants now appear to host *dilute* / *fuzzy* cores, not compact ones.
3. Saturn's spectacular rings are young (~100 Myr) and transient.
4. Galilean moons and Saturnian moons span unprecedented diversity: Io, Europa, Ganymede, Callisto, Titan, Enceladus.
5. Ice giants are the most under-explored major planets; one flyby each, ~35 yr ago.
6. Roche limit explains where rings exist and where moons cannot accrete:
 $\approx 2.46 R_p (\rho_p / \rho_s)^{1/3}$.



Questions?

Next: **Lecture 12. Small Bodies: Meteorites, Asteroids, Comets**

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience